

CodeAlpha Java Internship

Hotel Reservation System

Java Source Code

```
import java.util.*;

public class HotelReservationSystem {
    static class Room {
        int number;
        String category;
        double price;
        boolean booked;

        Room(int number, String category, double price) {
            this.number = number;
            this.category = category;
            this.price = price;
            this.booked = false;
        }
    }

    static class Booking {
        String guestName;
        int roomNumber;
        int nights;
        double amount;

        Booking(String guestName, int roomNumber, int nights, double amount) {
            this.guestName = guestName;
            this.roomNumber = roomNumber;
            this.nights = nights;
            this.amount = amount;
        }
    }

    static Scanner sc = new Scanner(System.in);
    static ArrayList<Room> rooms = new ArrayList<>();
    static ArrayList<Booking> bookings = new ArrayList<>();

    static void showRooms() {
        System.out.println("\n=== ROOM AVAILABILITY ===");
        System.out.printf("%-8s %-12s %-12s %-12s\n",
            "Room", "Category", "Price", "Status");

        for (Room r : rooms) {
            System.out.printf("%-8d %-12s %-12.2f %-12s\n",
                r.number, r.category, r.price,
                r.booked ? "Booked" : "Available");
        }
    }

    static void bookRoom() {
        showRooms();
        System.out.print("\nEnter room number: ");
        int number = sc.nextInt();

        Room selected = null;
        for (Room r : rooms)
            if (r.number == number) selected = r;

        if (selected == null) {
            System.out.println("Room not found.");
            return;
        }
        if (selected.booked) {
            System.out.println("Room is already booked.");
            return;
        }

        sc.nextLine();
        System.out.print("Guest name: ");
        String name = sc.nextLine();

        System.out.print("Number of nights: ");
        int nights = sc.nextInt();
        if (nights <= 0) {
            System.out.println("Nights must be positive.");
            return;
        }

        double amount = selected.price * nights;
        System.out.printf("Total amount: %.2f\n", amount);
        System.out.print("Confirm payment? (yes/no): ");
        String payment = sc.next();

        if (!payment.equalsIgnoreCase("yes")) {
```

```

        System.out.println("Booking cancelled.");
        return;
    }

    selected.booked = true;
    bookings.add(new Booking(name, number, nights, amount));
    System.out.println("Payment simulated successfully.");
    System.out.println("Room booked successfully.");
}

static void cancelBooking() {
    System.out.print("Enter room number to cancel: ");
    int number = sc.nextInt();

    Booking found = null;
    for (Booking b : bookings)
        if (b.roomNumber == number) found = b;

    if (found == null) {
        System.out.println("Booking not found.");
        return;
    }

    for (Room r : rooms)
        if (r.number == number) r.booked = false;

    bookings.remove(found);
    System.out.println("Booking cancelled successfully.");
}

static void showBookings() {
    System.out.println("\n=== BOOKING DETAILS ===");
    if (bookings.isEmpty()) {
        System.out.println("No bookings.");
        return;
    }

    for (Booking b : bookings) {
        System.out.printf("Guest: %s | Room: %d | Nights: %d | Amount: %.2F%n",
            b.guestName, b.roomNumber, b.nights, b.amount);
    }
}

public static void main(String[] args) {
    rooms.add(new Room(101, "Standard", 1500));
    rooms.add(new Room(102, "Standard", 1500));
    rooms.add(new Room(201, "Deluxe", 2500));
    rooms.add(new Room(202, "Deluxe", 2500));
    rooms.add(new Room(301, "Suite", 4000));

    int choice;
    do {
        System.out.println("\n=== HOTEL RESERVATION SYSTEM ===");
        System.out.println("1. Search/View Rooms");
        System.out.println("2. Book Room");
        System.out.println("3. Cancel Reservation");
        System.out.println("4. View Bookings");
        System.out.println("5. Exit");
        System.out.print("Choose: ");
        choice = sc.nextInt();

        switch (choice) {
            case 1 -> showRooms();
            case 2 -> bookRoom();
            case 3 -> cancelBooking();
            case 4 -> showBookings();
            case 5 -> System.out.println("Thank you.");
            default -> System.out.println("Invalid choice.");
        }
    } while (choice != 5);

    sc.close();
}
}

```

Project README

CodeAlpha Hotel Reservation System

A Java console project for searching, booking and cancelling hotel rooms.

Features

- Standard, Deluxe and Suite rooms
- Room availability
- Booking and cancellation
- Payment simulation
- Booking details view
- Uses OOP and ArrayList

Run

```
```bash
javac HotelReservationSystem.java
java HotelReservationSystem
```
```

Note

Payment is only simulated; this project does not process real money.